Anmol Sharma

AI Engineer | Python

Jaipur, India | +91-84128-80194 | ozymandias.work@gmail.com | LinkedIn | Github Portfolio

Python-focused AI Engineer with comprehensive experience in Generative AI, Classical Machine Learning and Deep Learning. Proven track record of building production-ready RAG pipelines, multi-agent systems, and financial forecasting models. Strong in latency optimization, evaluation, and end-to-end deployment with Python, FastAPI, Docker, and ONNX.

AREAS OF EXPERTISE

Generative AI: LLMs (GPT-4, Llama 3), RAG, Prompt Engineering, Multi-Agent Systems, LangChain
Machine Learning & CV: Ensemble Learning, Computer Vision (YOLO, OpenCV), Unsupervised Learning, NLP
Development & Operations: Python, SQL, FastAPI, Git, Docker, Linux, ONNX, Vector Databases

PROJECTS

Multi-Agent Personal Health Coach | Python, LangChain, RAG

- Designed a multi-agent wellness assistant that handled complex, multi-turn user intents and personalized guidance.
- Implemented hybrid RAG over structured medical data and articles, reducing hallucination rates by 40% and supporting voice interaction with sub-200ms latency.

Real-Time Face Mask Detection | Python, Computer Vision, YOLO

- Developed a real-time detection pipeline with YOLO/MobileNet to classify correct, incorrect, and missing face mask usage.
- Optimized CPU inference to ≥ 15 FPS with F1-score > 0.90 and deployed via ONNX Runtime for lightweight edge use.

Next-Day Stock Return Prediction | Python, Ensemble Learning, Time-Series

- Built a next-day return forecasting pipeline using a stacked ensemble of ElasticNet, Random Forest, and Gradient Boosting models.
- Implemented leakage-safe rolling features with walk-forward validation and market-context features to improve robustness in volatile periods.

PROFESSIONAL EXPERIENCE

Independent Research & Development — *June 2025 – Present*
AI Systems Architect — *Self-Directed / Open Source*

- Focused on designing production-ready AI systems with low-latency inference, agentic workflows, and robust state management.
- **Autonomous Agent Architecture (Nexus Bot):** Engineered a LangGraph-based multi-agent system with PostgreSQL-backed persistent memory for long-term contextual recall.
- **Neuro-Symbolic RAG System (MedPal AI):** Developed a clinical decision support system using Dual-RAG and specialized BERT models to handle domain-specific queries while reducing hallucination risk.
- **Real-Time Conversational AI:** Built an emotion-aware voice assistant with real-time intent recognition, Silero VAD, and Kokoro TTS for low-latency speech interaction.

Teleperformance — *Jaipur, India*
Technical Support Executive — *Jul 2023 – Apr 2024*

- Diagnosed software issues in production environments while maintaining a CSAT of 4.8/5.
- Translated user issues into technical requirements and used SQL-based analysis to improve tool reliability and decision-making.

EDUCATION

Indian Institute of Technology (IIT), Mandi — *May 2025 – Present*
Minor in Data Science and Artificial Intelligence

Jaipur National University — *Jun 2019 – May 2022*
Bachelor of Computer Applications (BCA)

HONORS & LANGUAGES

Honors: NTSE Scholar.
Languages: English (C2), Hindi (Native), Japanese (A2-B1).